

Securing Gesture Passwords Against Shoulder Surfing Using Behavioral Features

Eunyong Cheon
UNIST
Ulsan, Republic of Korea
beer@unist.ac.kr

Ian Oakley
School of Electrical Engineering
KAIST
Daejeon, Republic of Korea
ian.r.oakley@gmail.com

Abstract

Gestures drawn on touchscreens are an emerging smartphone authentication method with good usability and a theoretically large password space. However, they are highly susceptible to observation attacks. To address this, recent research has demonstrated the effectiveness of implicit authentication techniques and behavioral features in strengthening the security of knowledge-based passwords such as PINs or patterns. Building on these ideas, this late-breaking work explores the design of gesture password hardening techniques. We examine the impact of incorporating behavioral biometric features captured by commercial smartphone sensors to enhance existing gesture recognition techniques. We comprehensively evaluate our system in a controlled (N=20) study setting, showing it effectively resists video observation attacks (EER=3.56%). These results indicate that behavioral features can substantially improve the security of gesture passwords against this common threat.

CCS Concepts

• **Security and privacy** → **Graphical / visual passwords**; • **Human-centered computing** → *HCI design and evaluation methods*.

Keywords

Gesture password, Mobile authentication, Behavioral biometrics

ACM Reference Format:

Eunyong Cheon and Ian Oakley. 2025. Securing Gesture Passwords Against Shoulder Surfing Using Behavioral Features. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3706599.3720021>

1 Introduction

Smartphone security has become essential, as these devices provide critical services and store sensitive information in diverse areas such as communications, financial transactions, and health. To protect against malicious attackers who gain access to devices (e.g., via theft [16] or surreptitious use [22]), phone lock services are now very widely adopted [33]. While biometric authentication [1]

through techniques such as fingerprint [20] or facial ID [41] are popular due to their high perceived security and the low burdens they place on users [15], these schemes are still typically backed-up by a traditional knowledge-based credential, such as a PIN or pattern. This is used to formally authenticate users at key moments such as when a device is rebooted, updated, security settings are changed, when biometrics fail (e.g., due to skin abrasion or other physical damage to the fingertips [23]), or simply after a particular span of time (e.g., six days¹).

While the most common knowledge-based schemes on phones are PINs and patterns, various researchers have proposed and explored free-form gesture passwords—two-dimensional strokes and shapes drawn on a device’s touch screen—as authentication credentials due to the efficiency with which they can be entered [38], the reliability with which they can be recalled [38] and the large theoretical password space they embody [27]. However, despite their strong potential, gesture passwords, as a form of graphical password [12], are widely acknowledged to be susceptible to observation attacks [36] via in-person shoulder surfing [26] or remote video-recording [14]. One promising technique that may address this limitation and improve security against observation attacks is hardening gesture passwords with multi-layered authentication [21]. This approach has been applied to schemes such as PIN [42] and pattern [6] and involves collecting behavioral data from users’ devices as they enter their credentials and using this material as an implicit and complementary authentication system [6]. Essentially, such systems seek to match not just the user’s credentials but also key aspects of how those credentials are entered under the intuition that there will be substantial, meaningful, and hard-to-replicate variations in such data between individuals. Consequently, these techniques are considered particularly effective when a genuine user’s password has been leaked to, or observed by, an attacker [40]. While this idea is well-established in other contexts, its application to user-generated free-form gesture passwords, in terms of the compatibility of the behavioral features used with existing gesture recognition techniques, remains unexplored.

This paper addresses this omission and takes up this task: we seek to design and evaluate the effectiveness of a password hardening scheme for free-form gesture passwords. We achieve this by developing an implicit stroke feature-based classification system and layering it on top of a traditional gesture password-matching process that uses Dynamic Time Warping (DTW). While behavioral biometric features have been widely evaluated as standalone classification methods in prior work [6, 40], gesture-based authentication differs in that it does not rely on exact symbol matching (e.g.,

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3720021>

¹<https://support.apple.com/en-gb/guide/security/sec9479035f1/web>

as in pattern locks). Instead, distance metrics, such as DTW, are calculated to differentiate among user inputs. As such, rather than simply applying implicit behavioral features to an authentication system, our work enhances security by integrating these features with existing gesture recognition techniques, leveraging the complementary strengths of both modalities to improve authentication robustness. We evaluate the effectiveness of the combined scheme against observation attacks. To accomplish this, we first collected a large dataset comprising 10,630 gestures entered by 10 users. For each gesture, we recorded touch and motion sensor data. To create this dataset, each participant contributed 63 examples of their own user-generated gesture password (13 during the creation process, followed by an additional 50 repetitions). They then provided another 50 repetitions of 20 common gesture passwords taken from prior work [3]. Using this data, we developed per-user one-class classifiers based on a set of keystroke features (a subset of 137 generated from a literature review) and integrated them with the DTW recognizer. To assess the system's resilience, we conducted a follow-up lab study with 10 new participants performing video observation attacks on the user-generated gestures from the initial study. Results indicate that user-generated gestures were highly vulnerable if matched solely using the DTW recognizer, with attack success rates ranging between 90.4% and 96.4%. In contrast, our multilayered system demonstrated strong resilience to observation attacks by achieving substantially reduced attack success rates between 2.11% and 4.90% and promising Equal Error Rates (EERs) between 1.37% and 3.56%.

The contributions of this work lie in the design and evaluation of the hardening scheme for gesture passwords. We highlight the feasibility of integrating behavioral biometric classifiers with existing free-form gesture password recognition systems to effectively distinguish between users entering (or copying after observing) the same gestures. We document the performance of our multilayered system through two lab studies and against observation attacks. Our results demonstrate how gesture passwords can be secured against key threats by leveraging behavioral data. These findings address prior limitations of the technique and open it up to more widespread adoption in the future.

2 Related Work

Gesture Passwords. Gesture passwords have been proposed as a usable [27, 38], reliable [6, 27], and robust [4, 27] authentication method to safeguard users' private information on touchscreen devices. Gestures are typically evaluated using an empirically derived optimal threshold on metrics such as DTW [6, 31] or cosine [18] distance calculated between an entered gesture and a user's stored credential. Recent research has examined the security aspects of gesture passwords, challenging their efficacy against different threat models. This work has revealed that users generate highly guessable gesture passwords - Liu et al. [19], for example, conducted an offline attack on a gesture password dataset by composing large dictionaries of representative gestures. This approach successfully guessed between 47.71% and 55.9% of gesture passwords within 10^9 attempts. Similarly, Cheon et al. [4] proposed an online attack that used clustering on a large gesture password dataset to construct a dictionary capable of cracking between 54.18% and 58.37%

of gestures within 20 guesses. The high success rates of these attacks suggest that the theoretically large space of possible gesture passwords does not readily translate into users' selection of unique credentials.

Password Enhancements. To address security pitfalls and improve the security of gesture passwords, researchers have proposed techniques such as obfuscation [25, 27], password composition policies [4, 5] and strength measurement [3] to make users' input process difficult to observe or to motivate users to generate complex gesture passwords that attackers cannot easily guess. While such policies have been shown to be effective, these schemes often incur usability costs, such as reductions in recall rates and increases in setup time. While such costs may be acceptable given the benefits they provide, our work seeks to explore a complementary approach and determine whether hardening gesture passwords with behavioral features can provide similar or additional benefits with reduced costs in terms of usability.

Hardening Passwords. Knowledge-based authentication methods are widely accepted to be vulnerable to observation attacks [32] in which a nefarious third party gains access to information about a genuine user's credentials through the simple act of watching (in person, by shoulder surfing [26]) or remotely recording the password entry process [28]. A key way to tackle such risks is by hardening passwords with behavioral features extracted during the password entry process [21]. This approach integrates behavioral biometric features with a password to create a combined credential in which both the content users enter and the specifics of how they enter it are key factors during authentication. Various versions of this idea have been previously proposed for phone lock scenarios. For example, De Luca et al. [6] introduced a behavioral biometric element to pattern lock authentication that relied on touch sensor data and reported it could identify different users with 77% accuracy. However, such relatively simple schemes may still be susceptible to skilled attackers [14] who observe and closely mimic users' input processes. More recent work by Zhang et al. [40] suggests that combined features spanning both touch and inertial sensor data from smartphones may mitigate such concerns. This work uses multi-threshold kNN classifiers to achieve an Equal Error Rate (EER) of 4.90% for 20 common pattern locks and, additionally, demonstrates effective resistance to observation attacks (EERs ranging from 3.19% to 6.73%). Despite these promising results, we know of no prior work that has examined the potential of behavioral features to improve the security of gesture passwords. This work aims to address this omission.

3 Threat Model

This work considers a critical threat model for smartphone authentications: observation attack. We assume an attacker has not only access to a user's device but also access to clear, high-quality videos depicting multiple successful unlocks. In addition, attackers have sufficient time to repeatedly view this content and practice mimicking the genuine user's authentication input. Finally, they seek to enter these credentials on a user's device in a limited, fixed number of 20 attempts. We note these attacks are highly relevant for phone lock scenarios as such devices suffer elevated risks of theft compared to other platforms [35] and authentication attempts

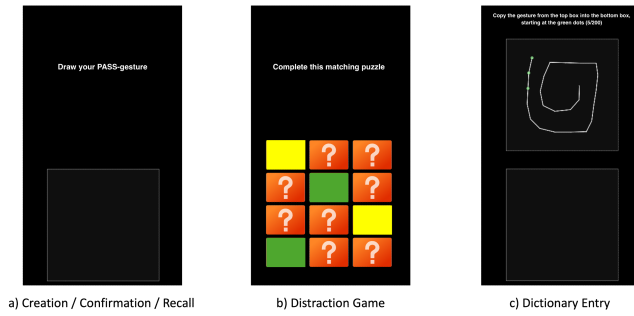


Figure 1: The screenshots depict the study procedures: (a) gesture password creation, confirmation, and recall; (b) distraction game; and (c) dictionary repetition phase.

(which are widely reported to be highly frequent) often take place in easy-to-observe public settings [8].

4 Multilayered Gesture Recognition

We first conducted a lab study with 10 users in a controlled, homogeneous setting (e.g., fixed device and pose) to characterize the security of gesture passwords against observation attacks and design implicit authentication methods that use behavioral features to improve their resistance. We achieved this by using behavioral data captured during the 10,630 gestures entered by our lab participants to develop classifiers capable of authenticating users by the specific and unique behaviors they exhibit during gesture entry. We explore key design aspects of the classifiers by exploring salient stroke features to be extracted from each gesture. Finally, we conducted an attack study to assess the performance of our system against observation attacks. By recording videos of genuine user participants entering their credentials, we created a data set for subsequent participants to view and use to conduct observation attacks, thus supporting an analysis of how effectively our system can resist such efforts. The local Institutional Review Board (IRB) reviewed and approved the study protocol.

4.1 Study Tasks and Procedure

The lab study for genuine users had two phases: the first involved them creating their own gesture passwords, and the second involved them repeatedly copying a representative set of existing gesture passwords.

During the first phase, participants were instructed to create a secure and memorable gesture password that could be used to lock their mobile phones (Figure 1, left). We enforced a simple policy for minimum gesture complexity: if participants created a trivial gesture (e.g., a single line, defined as a stroke with three or fewer points after Douglas-Puecker line simplification), they received a message to restart the gesture creation process. After successfully creating a gesture, participants confirmed it via entering it a second time. We verified this by matching gestures using Dynamic Time Warping (DTW) as a similarity metric with a threshold of 18.52, a figure reported in closely related prior work [3]. If participants failed to confirm their gesture, they restarted the gesture creation process. Otherwise, they progressed to the next stage of the study.

This involved practicing entering their gestures 10 times, receiving feedback each time regarding their attempt’s correctness. Next, they completed a color-matching puzzle game designed to distract them and occupy their short-term memory (Figure 1, middle). We selected this game based on its use in closely related prior work [3]. The game involved twelve colored cards organized into six same-colored pairs. All cards were initially flipped face-down. Users touched pairs of cards to reveal their colors, and when a pair of colors matched, the cards remained face up. Otherwise, they flipped over again, and the participant made a new pair of guesses. Once all six pairs of colors were matched, the puzzle was complete. Participants then recalled their gesture: they were given five attempts to re-enter their gesture password using the same criteria deployed when checking for password confirmation. As the final activity in this stage of the study, participants repeatedly entered their created gesture a further 50 times.

The study’s second phase involved participants repeatedly entering the full set of 20 gesture passwords in a dictionary of commonly generated gesture credentials reported in recent prior work [3]—see Figure 3. Specifically, participants entered each dictionary gesture 50 times (so making 1,000 entries in total). This task was divided into five equally sized blocks of 200 gesture entries, each separated by a short break (at least one minute) to reduce fatigue. Within each block, each gesture was entered ten times, and the order of gesture entry was fully randomized. We ensured faithful entry of the dictionary gestures through two mechanisms. First, we presented the target gesture to be entered on the screen above the gesture input area, with the start point and initial stroke direction highlighted—see Figure 1, Right. After each gesture entry, we matched the user-entered gesture against the dictionary original, using the same DTW metric and threshold as during gesture creation; exceeding the distance metric threshold meant the gesture needed to be re-entered. All tasks were completed using the same smartphone (iPhone 15 Pro) we provided. Participants were requested to maintain the same posture throughout the session: they sat on a chair, held the phone in their non-dominant hand, and rested their elbows on a desk in front of them with their forearms at approximately 45 degrees from the horizontal (i.e., with the phone around 20cm above the desk). We collected consent and the information necessary for providing compensation using paper forms. The mean duration of the genuine user study was 27 minutes, and participants were compensated with approximately \$22 USD for their participation.

The attack study was conducted independently from the genuine user study and involved a different set of participants. Attackers were invited to the lab and instructed to guess each genuine user’s gesture password within 10 attempts after watching videos of their password entry process (Figure 2). We repeated the set of attacks under two scenarios—naïve and informed—meaning that each attacker performed a total of 200 attacks, corresponding to 20 attacks on each genuine user. In the initial naïve scenario, attackers attempted to guess the genuine gestures solely by observing the videos, with no additional information provided. In the subsequent informed scenario, attackers were given specific guidelines about how the system functions and key aspects of genuine users’ behaviors to take note of, including details of drawing speed, the hand holding the device, the hand and finger used for entering the gesture, and

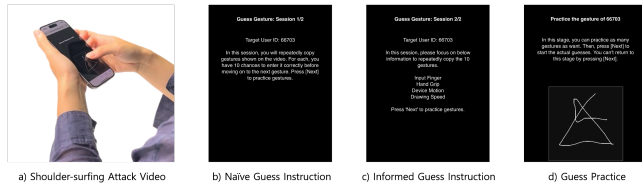


Figure 2: The figure illustrates a) the video footage shown to attackers, b) the instructions screen for Naive attacks, c) informed attacks, and d) the practice stage in the attack study.

the overall body posture. Before conducting an attack, attackers could replay and practice the genuine user gestures as many times as they wanted. The average study duration was 24 minutes, and participants were compensated \$15 USD upon completion of the study. To motivate the attackers, we offered a bonus (of \$10 USD) for each successfully guessed gesture.

4.2 Participants

We recruited a total of 10 participants for both the genuine user and attack studies, with five male and five female participants completing each study. All participants were Asian, and the majority (except for one left-handed individual in the genuine user study) were right-handed. Most participants (six in the genuine user study and nine in the attack study) were between 18-24 years of age, while the remainder were between 24-34 years. They had either completed high school (seven and nine participants in the genuine user and attack studies, respectively) or a college degree. They were majoring in electrical engineering, computer science or biology.

4.3 Classifier Design

Gesture Segmentation. The literature suggests that free-form gesture passwords are best modeled as a series of concatenated sub-strokes [4] that can feature (or be delimited by) segments involving either continuous motion or short pauses (or abrupt changes in direction). Prior literature also indicates that the features contributed by these different types of stroke element may also exhibit contrasting properties, and recording feature data specific to each may better support effective authentication [40]. Reflecting these observations, we designed segmentation algorithms to separate gestures into *stroke-segments* (representing motions) and *node-segments* (representing pauses, changes in direction, or similar inflection points). Furthermore, as segmentation algorithms typically precisely identify node-segments at specific, individual stroke points, we manually expanded each node-segment we identified to encompass surrounding stroke points (defined as 5% of gesture length). This ensured that each node-segment represented the movements around a key change in stroke execution and, further, that it was represented by sufficient data from which to calculate behavioral features. The intuition behind our chosen segmentation technique is simple: relevant sub-strokes within a gesture will be delimited by brief pauses followed by rapid motions that mark the initial movements in a sub-stroke. To identify such moments, we calculate the speed between all successive points, apply a low-pass filter with a cutoff frequency of 5% of the sample rate to eliminate noise, and look for

local minima. We labeled these minima, representing the transitions between sub-strokes, as node-segments.

Stroke Features. We compiled and evaluated a comprehensive range of stroke features drawn from multiple sources, including research on implicit mobile authentication [6, 40], handwritten signature analysis [9], behavioral stroke features [10, 30], and features used in training probabilistic models [3]. We categorized these features into three types: *input* features related to the act of drawing the gesture password on the screen, *shape* features describing the gesture itself, and *motion* features representing the movements of the device made while drawing the gesture. In the following sections, we provide a more detailed description.

Input Features. We include seven features based on the entry of an entire gesture. These were the time taken (ms) [10], the mean drawing speed (pixels/s) [30], the number of velocity peaks in the gesture [9], and the normalized x and y coordinates of the start and end positions [40]. We also included 16 features describing the different segments recovered from the gesture. Specifically, we calculate four summary statistics (mean, SD, max, and min) for segment size (or number of points) and speed (m/s) over the sets of both stroke- and node-segments. This contributes a further 16 features, leading to a total of 23 input features.

Shape Features. We calculated six features describing the overall form of an entered gesture. These were its proportion (the ratio between its width and height) [13], its displacement [30] and direction [24] (respectively, the distance between start and end points and the angle from start to end points), its distance (or overall length) [24], the number of strokes it was composed of [2] and its curvature (measured by angle) [37]. We included an additional 12 shape features relating to the sets of gesture segments, again in terms of four summary statistics (mean, SD, max, min). Specifically, we calculated these summary statistics for the curvature of both stroke- and node-segments and for the symmetry of stroke-segments [3]. Symmetry was assessed by dividing each stroke-segment into a pair of sub-strokes centered on the stroke-segment's midpoint and then matching these with DTW. Calculating symmetry for node-segments was considered meaningless due to their small physical size.

Motion Features. We calculated features from the IMU data collected during gesture entry. This dataset includes eight separate data series: tri-axial measurements of linear acceleration [11] (which encompass gravitational forces) and rotational speed [29], and core motion metrics capturing device pitch and roll [17]. Features were again based on the same four summary statistics (mean, SD, min, max) calculated over each of these eight data series for the whole gesture (32 features in total) and for the sets of both stroke- and node segments (a further 64 features). Thus, we included a total of 96 motion features.

Feature Selection. We calculated Fisher Scores [34], or the ratio of the variance between different users over the variance within individual users, to select the subset of 25 features (from our initial set of 137) that most effectively discriminate between participants. To achieve this, we calculated mean Fisher Scores for all features on all dictionary gestures (Figure 3). Only those features with Fisher Scores exceeding a threshold value of 1 were included in the final set of features used during classifier development. Appendix A contains the full list of all features retained through this process.

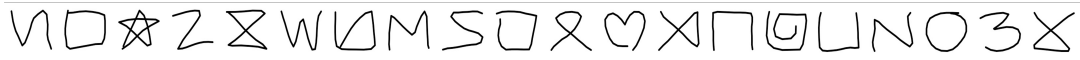


Figure 3: 20 dictionary gesture passwords evaluated in this study.

4.4 Security Results

We performed a security analysis using a one-class Support Vector Machine (SVM) classifier trained on a selected subset of stroke features extracted from dictionary gestures. The idea was to identify stroke features capable of distinguishing identical gesture inputs across different users, providing a robust security layer for user-generated gestures against observation attacks.

Evaluation Setup. For our one-class classifier, we ran a five-fold cross-validation procedure in which 80% (40 gestures) of the data from the repeated entry of their own created gesture was used for training and the remaining 20% (10 gestures) for genuine user testing. The imposter test set was composed of the relevant gesture inputs contributed by all ten attackers, resulting in a substantial number of attacks: 100 in total. We report results in terms of the mean EER over all participants.

Observation Attack Result. We first calculated the attack success rate using a standard DTW-based gesture password system. The results were stark—the naive attack was successful in 90.40% of attempts, while the subsequent informed attack succeeded 96.4% of the time. This upward trend may be due to the additional practice attackers gained as they went through the study or reflect the information provided to them about the operation of the system before conducting the informed attacks. Regardless, these results highlight the ease with which video observation can compromise gesture passwords—if attackers can record a standard gesture password, they can readily replicate it. We then applied the multi-layer authentication scheme proposed in this work. Specifically, after using DTW to successfully match an entered gesture to a user’s stored credential, we submitted behavioral features captured during the input process to the appropriate one-class classifier. This led to a substantially reduced number of compromised passwords: attack success rates were significantly reduced to between 2.11% (naive attack scenario) and 4.90% (informed attackers). The corresponding EERs for these attacks remained very low, ranging from 1.37% to 3.56%. These results suggest that compromising gesture passwords hardened with behavioral features is a challenging task, even when provided with a high-quality video depicting both the gestures and the gesture entry processes.

5 Limitation and Future Work

This work proposed, implemented, and evaluated a password hardening approach to improving the security of gesture-based credentials with behavioral features captured during authentication. We first conducted a lab study to support classifier development. We then conducted a follow-up study to attack the collected credentials. Our multilayered gesture authentication system demonstrated good resistance to observation attacks. We discuss the implications of these findings in detail below.

While breach rates were very high for standard DTW recognizers (between 90.40% and 96.50%), indicating that observed gestures are

very readily compromised, adopting the multi-layered scheme proposed in this work led to dramatically reduced figures of between 2.11% and 4.90%. Based on these strong improvements, we argue that our approach may be more practical than prior proposals in this area, which have tended to focus on simply obfuscating input processes using techniques such as magnetic input [25], multi-finger gestures [27], interface modulations [39], and increased input dimensions [7]. It is also worth comparing the security improvements achieved in this work to implicit authentication schemes for other knowledge-based password systems. For example, in closely related work exploring the design of classifiers to harden pattern lock input, one-class EER results for typical observation attack conditions are 4.92%, with more severe conditions yielding an EER of 6.55% [40]. The lower figures attained in this work – one-class EERs between 1.37% to 3.56% suggest that the rich, expressive gesture passwords studied in this work may be better protected by authentication classifiers leveraging behavioral features than the simpler inputs involved in entering pattern locks.

There are several other limitations to our work and many directions for further study. Future work on this topic should seek to capture larger data sets of user-generated gestures to support more extensive comparisons with prior work. Furthermore, our feature and classifier selection work could be extended in many ways. For example, we limited sample rates in our work to approximately 60Hz, whereas higher rates may have enabled more effective segmentation techniques and higher fidelity features; we omitted hyper-parameter selection procedures to avoid over-fitting; our set of four basic summary features (mean, SD, min, max) could be easily extended. Additionally, to enhance the real-world applicability of our classification system, future work should evaluate whether our model can maintain performance over longitudinal login attempts and adapt to variations in entry poses. Future work should explore these variations. More speculatively, it would also be interesting to study the performance of the behavioral classifiers proposed in this paper when combined with alternative distance metrics, such as cosine distance [18], or more complex gesture input approaches, such as those based on multi-finger, multi-stroke techniques [38].

6 Conclusion

This work sought to develop a password hardening technique for gesture passwords. We achieved this by exploring behavioral features can effectively support distinguishing between users as they enter gesture passwords and by training classification models that can serve as an implicit behavioral biometric authentication layer on top of existing gesture password authentication schemes. Results indicate our approach effectively resists observation attacks. In this way, our work represents a meaningful step towards realizing gesture passwords as a viable authentication techniques that can provide strong security for its users and help secure their devices against attackers, while ensuring that genuine users are presented with few barriers when accessing their devices.

Acknowledgments

This work was supported by the Basic Science Research Program through the NRF funded by the Ministry of Science, ICT and Future Planning (2023R1A2C1004046).

References

- [1] FIDO Alliance. 2023. Favorite security authentication methods in selected countries 2023. <https://www.statista.com/statistics/1448883/preferred-security-authentication-methods-in-selected-countries/>.
- [2] Lisa Anthony, Radu-Daniel Vatavu, and Jacob O. Wobbrock. 2013. Understanding the consistency of users' pen and finger stroke gesture articulation. In *Proceedings of Graphics Interface 2013* (Regina, Saskatchewan, Canada) (GI '13). Canadian Information Processing Society, CAN, 87–94.
- [3] Eunyong Cheon, Jun Ho Huh, and Ian Oakley. 2023. GestureMeter: Design and Evaluation of a Gesture Password Strength Meter. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (<conf-loc>, <city>Hamburg</city>, <country>Germany</country>, </conf-loc>) (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 69, 19 pages. <https://doi.org/10.1145/3544548.3581397>
- [4] Eunyong Cheon, Yonghwan Shin, Jun Ho Huh, Hyounghshick Kim, and Ian Oakley. 2020. Gesture authentication for smartphones: Evaluation of gesture password selection policies. In *2020 IEEE Symposium on Security and Privacy (SP)*. IEEE, IEEE, San Francisco, CA, USA, 249–267.
- [5] Gradeigh D Clark, Janne Lindqvist, and Antti Oulasvirta. 2017. Composition policies for gesture passwords: User choice, security, usability and memorability. In *2017 IEEE Conference on Communications and Network Security (CNS)*. IEEE, IEEE, Las Vegas, NV, USA, 1–9.
- [6] Alexander De Luca, Alina Hang, Frederik Brudy, Christian Lindner, and Heinrich Hussmann. 2012. Touch Me Once and I Know It's You! Implicit Authentication Based on Touch Screen Patterns. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Austin, Texas, USA) (CHI '12). Association for Computing Machinery, New York, NY, USA, 987–996. <https://doi.org/10.1145/2207676.2208544>
- [7] Alexander De Luca, Emanuel Von Zezschwitz, Ngo Dieu Huong Nguyen, Max-Emanuel Maurer, Elisa Rubegni, Marcello Paolo Scipioni, and Marc Langheinrich. 2013. Back-of-device authentication on smartphones. In *Proceedings of the sigchi conference on human factors in computing systems*. Association for Computing Machinery, New York, NY, USA, 2389–2398.
- [8] Malin Eiband, Mohamed Khamis, Emanuel Von Zezschwitz, Heinrich Hussmann, and Florian Alt. 2017. Understanding shoulder surfing in the wild: Stories from users and observers. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA, 4254–4265.
- [9] Julian Fierrez-Aguilar, Loris Nanni, Jaime Lopez-Penalba, Javier Ortega-Garcia, and Davide Maltoni. 2005. An on-line signature verification system based on fusion of local and global information. In *Audio and Video-Based Biometric Person Authentication: 5th International Conference, AVBPA 2005, Hilton Rye Town, NY, USA, July 20-22, 2005. Proceedings 5*. Springer, Springer, Berlin, Heidelberg, 523–532.
- [10] Mario Frank, Ralf Biedert, Eugene Ma, Ivan Martinovic, and Dawn Song. 2012. Touchalytics: On the applicability of touchscreen input as a behavioral biometric for continuous authentication. *IEEE transactions on information forensics and security* 8, 1 (2012), 136–148.
- [11] Javier Guerra-Casanova, Carmen Sánchez-Ávila, Gonzalo Bailador, and Alberto de Santos Sierra. 2012. Authentication in mobile devices through hand gesture recognition. *International Journal of Information Security* 11 (2012), 65–83.
- [12] Ian Jermyn, Alain Mayer, Fabian Monrose, Michael K Reiter, and Avi Rubin. 1999. The design and analysis of graphical passwords. In *8th USENIX Security Symposium (USENIX Security 99)*. USENIX Association, Washington, D.C.
- [13] Shaun K. Kane, Jacob O. Wobbrock, and Richard E. Ladner. 2011. Usable gestures for blind people: understanding preference and performance. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Vancouver, BC, Canada) (CHI '11). Association for Computing Machinery, New York, NY, USA, 413–422. <https://doi.org/10.1145/1978942.1979001>
- [14] Hassan Khan, Urs Hengartner, and Daniel Vogel. 2016. Targeted Mimicry Attacks on Touch Input Based Implicit Authentication Schemes. In *Proceedings of the 14th Annual International Conference on Mobile Systems, Applications, and Services* (Singapore, Singapore) (MobiSys '16). Association for Computing Machinery, New York, NY, USA, 387–398. <https://doi.org/10.1145/2906388.2906404>
- [15] Masoud Mehrabi Koushki, Borke Obada-Obieh, Jun Ho Huh, and Konstantin Beznosov. 2021. On Smartphone Users' Difficulty with Understanding Implicit Authentication. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 690, 14 pages. <https://doi.org/10.1145/3411764.3445386>
- [16] Tao Li, Yimin Chen, Jingchao Sun, Xiaocong Jin, and Yanchao Zhang. 2016. iLock: Immediate and Automatic Locking of Mobile Devices against Data Theft. In *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security* (Vienna, Austria) (CCS '16). Association for Computing Machinery, New York, NY, USA, 933–944. <https://doi.org/10.1145/2976749.2978294>
- [17] Wen-Jeng Li, Chia-Yeh Hsieh, Li-Fong Lin, and Woei-Chyn Chu. 2017. Hand gesture recognition for post-stroke rehabilitation using leap motion. In *2017 international conference on applied system innovation (ICASI)*. IEEE, IEEE, Sapporo, Japan, 386–388.
- [18] Yang Li. 2010. Protractor: a fast and accurate gesture recognizer. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Atlanta, Georgia, USA) (CHI '10). Association for Computing Machinery, New York, NY, USA, 2169–2172. <https://doi.org/10.1145/1753326.1753654>
- [19] Can Liu, Gradeigh D. Clark, and Janne Lindqvist. 2017. Guessing Attacks on User-Generated Gesture Passwords. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 1, 1, Article 3 (mar 2017), 24 pages. <https://doi.org/10.1145/3053331>
- [20] Davide Maltoni, Dario Maio, Anil K Jain, Salil Prabhakar, et al. 2009. *Handbook of fingerprint recognition*. Vol. 2. Springer, London.
- [21] Fabian Monrose, Michael K. Reiter, and Susanne Wetzel. 1999. Password hardening based on keystroke dynamics. In *Proceedings of the 6th ACM Conference on Computer and Communications Security* (Kent Ridge Digital Labs, Singapore) (CCS '99). Association for Computing Machinery, New York, NY, USA, 73–82. <https://doi.org/10.1145/319709.319720>
- [22] Fabio Pierazzi, Ghita Mezzour, Qian Han, Michele Colajanni, and V. S. Subrahmanian. 2020. A Data-driven Characterization of Modern Android Spyware. *ACM Trans. Manage. Inf. Syst.* 11, 1, Article 4 (apr 2020), 38 pages. <https://doi.org/10.1145/3382158>
- [23] Lina Qiu, Alexander De Luca, Ildar Muslukhov, and Konstantin Beznosov. 2019. Towards Understanding the Link Between Age and Smartphone Authentication. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland Uk) (CHI '19). Association for Computing Machinery, New York, NY, USA, 1–10. <https://doi.org/10.1145/3290605.3300393>
- [24] Dean Rubine. 1991. Specifying gestures by example. *ACM SIGGRAPH computer graphics* 25, 4 (1991), 329–337.
- [25] Alireza Sahami Shirazi, Peyman Moghadam, Hamed Ketabdar, and Albrecht Schmidt. 2012. Assessing the Vulnerability of Magnetic Gestural Authentication to Video-Based Shoulder Surfing Attacks. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Austin, Texas, USA) (CHI '12). Association for Computing Machinery, New York, NY, USA, 2045–2048. <https://doi.org/10.1145/2207676.2208352>
- [26] Florian Schaub, Marcel Walch, Bastian Könings, and Michael Weber. 2013. Exploring the design space of graphical passwords on smartphones. In *Proceedings of the Ninth Symposium on Usable Privacy and Security* (Newcastle, United Kingdom) (SOUPS '13). Association for Computing Machinery, New York, NY, USA, Article 11, 14 pages. <https://doi.org/10.1145/2501604.2501615>
- [27] Michael Sherman, Gradeigh Clark, Yulong Yang, Shridatt Sugrim, Arttu Modig, Janne Lindqvist, Antti Oulasvirta, and Teemu Roos. 2014. User-Generated Free-Form Gestures for Authentication: Security and Memorability. In *Proceedings of the 12th Annual International Conference on Mobile Systems, Applications, and Services* (Bretton Woods, New Hampshire, USA) (MobiSys '14). Association for Computing Machinery, New York, NY, USA, 176–189. <https://doi.org/10.1145/2594368.2594375>
- [28] Peipei Shi, Bo Zhu, and Amr Youssef. 2009. A PIN entry scheme resistant to recording-based shoulder-surfing. In *2009 Third International Conference on Emerging Security Information, Systems and Technologies*. IEEE, IEEE, Athens, Greece, 237–241.
- [29] Zdenka Sitová, Jaroslav Šeděnka, Qing Yang, Ge Peng, Gang Zhou, Paolo Gasti, and Kiran S Balagani. 2015. HMOG: New behavioral biometric features for continuous authentication of smartphone users. *IEEE Transactions on Information Forensics and Security* 11, 5 (2015), 877–892.
- [30] Yunpeng Song, Zhongmin Cai, and Zhi-Li Zhang. 2017. Multi-touch authentication using hand geometry and behavioral information. In *2017 IEEE symposium on security and privacy (SP)*. IEEE, IEEE, San Jose, CA, USA, 357–372.
- [31] Eugene M. Taranta II, Amirreza Samiei, Mehran Maghousi, Pooya Khallou, Corey R. Pittman, and Joseph J. LaViola Jr. 2017. Jackknife: A Reliable Recognizer with Few Samples and Many Modalities. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* (Denver, Colorado, USA) (CHI '17). Association for Computing Machinery, New York, NY, USA, 5850–5861. <https://doi.org/10.1145/3025453.3026002>
- [32] Furkan Tari, A. Ant Ozok, and Stephen H. Holden. 2006. A comparison of perceived and real shoulder-surfing risks between alphanumeric and graphical passwords. In *Proceedings of the Second Symposium on Usable Privacy and Security* (Pittsburgh, Pennsylvania, USA) (SOUPS '06). Association for Computing Machinery, New York, NY, USA, 56–66. <https://doi.org/10.1145/1143120.1143128>
- [33] Dirk Van Bruggen, Shu Liu, Mitch Kajzer, Aaron Striegel, Charles R. Crowell, and John D'Arcy. 2013. Modifying smartphone user locking behavior. In *Proceedings of the Ninth Symposium on Usable Privacy and Security* (Newcastle, United Kingdom) (SOUPS '13). Association for Computing Machinery, New York, NY, USA, Article

- 10, 14 pages. <https://doi.org/10.1145/2501604.2501614>
- [34] Sa Wang, Cheng-Lin Liu, and Lian Zheng. 2007. Feature selection by combining fisher criterion and principal feature analysis. In *2007 International Conference on Machine Learning and Cybernetics*, Vol. 2. IEEE, IEEE, Hong Kong, China, 1149–1154.
 - [35] Dan Whitworth. 2023. Mobile fraud: Thieves 'shoulder surfing' victims to steal phones. <https://www.bbc.com/news/business-65456325>. Accessed: 2024-09-09.
 - [36] Susan Wiedenbeck, Jim Waters, Leonardo Sobrado, and Jean-Camille Birget. 2006. Design and evaluation of a shoulder-surfing resistant graphical password scheme. In *Proceedings of the working conference on Advanced visual interfaces*. Association for Computing Machinery, New York, NY, USA, 177–184.
 - [37] Don Willems, Ralph Niels, Marcel van Gerven, and Louis Vuurpijl. 2009. Iconic and multi-stroke gesture recognition. *Pattern recognition* 42, 12 (2009), 3303–3312.
 - [38] Yulong Yang, Gradeigh D. Clark, Janne Lindqvist, and Antti Oulasvirta. 2016. Free-Form Gesture Authentication in the Wild. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems* (San Jose, California, USA) (CHI '16). Association for Computing Machinery, New York, NY, USA, 3722–3735. <https://doi.org/10.1145/2858036.2858270>
 - [39] Nur Haryani Zakaria, David Griffiths, Sacha Brostoff, and Jeff Yan. 2011. Shoulder surfing defence for recall-based graphical passwords. In *Proceedings of the Seventh Symposium on Usable Privacy and Security* (Pittsburgh, Pennsylvania) (SOUPS '11). Association for Computing Machinery, New York, NY, USA, Article 6, 12 pages. <https://doi.org/10.1145/2078827.2078835>
 - [40] Xincheng Zhang, Yafeng Yin, Lei Xie, Hao Zhang, Zefan Ge, and Sanglu Lu. 2020. TouchID: User Authentication on Mobile Devices via Inertial-Touch Gesture Analysis. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 4, 4, Article 162 (dec 2020), 29 pages. <https://doi.org/10.1145/3432192>
 - [41] Wenyi Zhao, Rama Chellappa, P Jonathon Phillips, and Azriel Rosenfeld. 2003. Face recognition: A literature survey. *ACM computing surveys (CSUR)* 35, 4 (2003), 399–458.
 - [42] Nan Zheng, Kun Bai, Hai Huang, and Haining Wang. 2014. You are how you touch: User verification on smartphones via tapping behaviors. In *2014 IEEE 22nd International Conference on Network Protocols*. IEEE, IEEE, Raleigh, NC, USA, 221–232.

A Appendix A

Table 1: Selected features for the granularity of gesture password, presented by their names, Fisher scores and counts.

Granularity	Selected Features	Fisher Score	Number of Selected Features
Overall	Max Pitch	2.50	13
	SD Rotation X	2.35	
	Mean Pitch	2.27	
	Min Pitch	1.73	
	SD Pitch	1.46	
	SD Rotation Y	1.45	
	Distance	1.42	
	Min Rotation Y	1.31	
	SD Rotation Z	1.08	
	Max Rotation X	1.07	
	SD Roll	1.05	
	Touch End Position Y	1.04	
	Mean Rotation X	1.02	
Node	Mean Pitch	1.50	4
	Mean Acceleration X	1.25	
	Min Pitch	1.14	
	Max Pitch	1.13	
Stroke	Mean Pitch	1.96	8
	Mean Acceleration X	1.87	
	Max Pitch	1.78	
	Min Pitch	1.77	
	Min Acceleration X	1.70	
	Max Acceleration X	1.70	
	Min Rotation X	1.13	
	SD Rotation X	1.02	